Database Languages

DDL, DML, and SQL Notes

1.4 Database Languages

Definition

A database system provides:

1. **Data-Definition Language (DDL):** Specifies the database schema.
2. **Data-Manipulation Language (DML):** Expresses queries and updates.

In practice, DDL and DML are integrated into a single language such as SQL.

1.4.1 Data Definition Language (DDL)

Definition

DDL allows users to:

1. Specify **data storage and definition** (e.g., B+ trees, hash indexing, sequential access).
2. Define **consistency constraints** for data integrity.
3. Manage **authorization** for different users and actions (read, insert, update, delete, create, alter, drop, rename).

Note

Consistency constraints include:

- **Domain constraints:** restrict attribute values to a specific domain (integer, char, date, etc.).
- **Referential integrity:** ensures values in one relation exist in another.
- **Authorization:** defines permissible operations for users.

Data Dictionary

Definition

Processing DDL statements updates the **data dictionary**, a special set of tables storing metadata about:

- **Relation-metadata:** relation name, number of attributes, storage, location.
- **Attribute-metadata:** attribute name, relation name, domain, position, length.
- **User-metadata:** user name, encrypted password, groups.
- **Index-metadata:** index name, relation name, type, attributes.
- **View-metadata:** view name and definition.

SQL Data Definition Language (DDL)

Example

Example SQL DDL to create an instructor table:

```
CREATE TABLE instructor (
    ID CHAR(5),
    name VARCHAR(20) NOT NULL,
    dept_name VARCHAR(20) NOT NULL,
    salary NUMERIC(8,2) CHECK (salary > 29000),
    PRIMARY KEY (ID),
    FOREIGN KEY (dept_name) REFERENCES department(dept_name)
        ON DELETE SET NULL
);
```

1.4.3 Data Manipulation Language (DML)

Definition

DML allows users to access or manipulate data:

- **Insert:** Add new records.
- **Delete:** Remove records.
- **Update:** Modify existing records.
- **Query:** Retrieve information.

Types of DML:

1. **Procedural DML:** Specify what data and how to retrieve.
2. **Declarative DML:** Specify what data is needed (e.g., SQL).

SQL Query Language

Example

Find all instructors in the Comp. Sci. department:

```
SELECT name
FROM instructor
WHERE dept_name = 'Comp. Sci.';
```

Find instructor ID and department name for departments with budget > 95000:

```
SELECT instructor.ID, department.dept_name
FROM instructor, department
WHERE instructor.dept_name = department.dept_name
AND department.budget > 95000;
```

Database Access from Application Programs

Definition

Non-procedural query languages (SQL) cannot perform:

- User input/output
- Network communication

Such tasks are implemented in host programming languages (C/C++, Java, Python) with embedded SQL.

Note

Standards for application program access:

- **ODBC:** C and other languages interface to databases.
- **JDBC:** Java interface to databases.

Applications can interact via SQL extensions or APIs like ODBC/JDBC.